

Edge-Magic Total Labelings on a Four-Partite Graph Family $G(m, n, k, t)$

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Abstract

An *edge-magic total labeling* (EMTL) of a finite simple graph $G = (V, E)$ is a bijection $f: V \cup E \rightarrow \{1, 2, \dots, |V| + |E|\}$ for which there is a constant κ with $f(u) + f(uv) + f(v) = \kappa$ for every edge $uv \in E$. Deciding whether an arbitrary graph admits such a labeling is computationally hard, and explicit constructions are known only for restricted graph classes. This paper studies EMTL on a parameterized family of four-partite graphs $G(m, n, k, t)$, built from two complete-bipartite layers joined by a circulant t -regular middle layer. We formalize the family, derive a degree-weighted identity that every EMTL must satisfy together with a divisibility condition and universal bounds on the magic constant, and prove a *vertex-determination* reduction showing that a labeling is fixed once the vertex labels and κ are chosen. Building on these facts, we cast EMTL existence as a constraint-satisfaction problem and solve it exactly with the OR-Tools CP-SAT solver, which either returns a verified labeling, a certificate of infeasibility, or reports a timeout. We report experiments across a range of parameters and provide an open-source reference implementation with a test suite and visualizer.

Keywords: edge-magic total labeling, graph labeling, constraint programming, CP-SAT, four-partite graphs.

MSC 2020: 05C78 (graph labeling); 68R10 (graph theory in computer science).

1 Introduction

A *graph labeling* assigns integers to the vertices and/or edges of a graph subject to structural conditions; the area originates with the work of Kotzig and Rosa on *magic valuations* [1], and the many variants studied since are catalogued in Gallian’s dynamic survey [3] and Wallis’ monograph [2]. In an *edge-magic total labeling* every integer in $\{1, \dots, |V| + |E|\}$ is used exactly once across all vertices and edges (a global bijection), while every edge is forced to the same induced sum (a local equation). The interaction of these two requirements is what makes the problem delicate: deciding existence is intractable in general, and most positive results take the form of explicit constructions for particular graph classes such as cycles, paths, wheels, and books [1, 3, 4, 6].

This paper takes a complementary, computational, and family-specific viewpoint. We fix a structured family of four-partite graphs $G(m, n, k, t)$ that is rich enough to be interesting yet regular enough to support systematic experimentation, and we attack EMTL existence on this family *exactly* using constraint programming. Concretely, the contributions are:

- (i) a precise definition of the graph family $G(m, n, k, t)$ and its basic structural invariants (Section 3);

- (ii) algebraic *necessary conditions* for an EMTL of $G(m, n, k, t)$ — a degree-weighted identity, a divisibility/congruence condition, and universal bounds on the magic constant (Section 4);
- (iii) a *vertex-determination* reduction: once the magic constant and the vertex labels are chosen, all edge labels are forced (Proposition 5), which both clarifies the search space and tightens the constraint model (Section 5);
- (iv) an exact CP-SAT formulation and an open-source reference implementation, together with an experimental study across parameter regimes (Sections 5–7).

The necessary conditions are not used to *decide* instances — the solver does that — but they serve as solver bounds and as independent post-solve correctness invariants, and they explain why the magic constant cannot be chosen freely.

2 Preliminaries

Throughout, $G = (V, E)$ is a finite simple undirected graph, $N := |V| + |E|$, and $\deg(v)$ denotes the degree of a vertex v .

Definition 1 (Total labeling). A *total labeling* of G is a function $f: V \cup E \rightarrow \{1, 2, \dots, N\}$. It is *bijective* if it is one-to-one and onto; equivalently, the multiset of assigned labels is exactly $\{1, 2, \dots, N\}$.

Definition 2 (Edge-magic total labeling). A bijective total labeling f is an *edge-magic total labeling* (EMTL) if there is a constant $\kappa \in \mathbb{Z}$ such that

$$f(u) + f(uv) + f(v) = \kappa \quad \text{for every edge } uv \in E.$$

The constant κ is the *magic constant* (or *magic sum*). A graph that admits an EMTL is called *edge-magic*.

Verification is cheap: given a candidate (κ, f) one checks bijectivity and the $|E|$ edge equations in $O(N)$ time. Finding a labeling, or certifying that none exists, is the hard direction and is the object of this paper.

3 The graph family $G(m, n, k, t)$

3.1 Vertex partitions and edge layers

Fix integers $m, n, k \geq 1$ and an integer t with $0 \leq t \leq n$. The vertex set is partitioned into four disjoint parts

$$A = \{A_0, \dots, A_{m-1}\}, \quad B = \{B_0, \dots, B_{n-1}\}, \quad C = \{C_0, \dots, C_{n-1}\}, \quad D = \{D_0, \dots, D_{k-1}\},$$

with $|A| = m$, $|B| = |C| = n$, $|D| = k$, so that

$$|V| = m + 2n + k. \tag{1}$$

The edge set is the union of three bipartite layers:

- (1) the *left* complete-bipartite layer $K_{m,n}$ between A and B (all mn edges);
- (2) the *middle* t -regular bipartite layer between B and C (nt edges);
- (3) the *right* complete-bipartite layer $K_{n,k}$ between C and D (all nk edges).

Hence

$$|E| = mn + nt + nk. \tag{2}$$

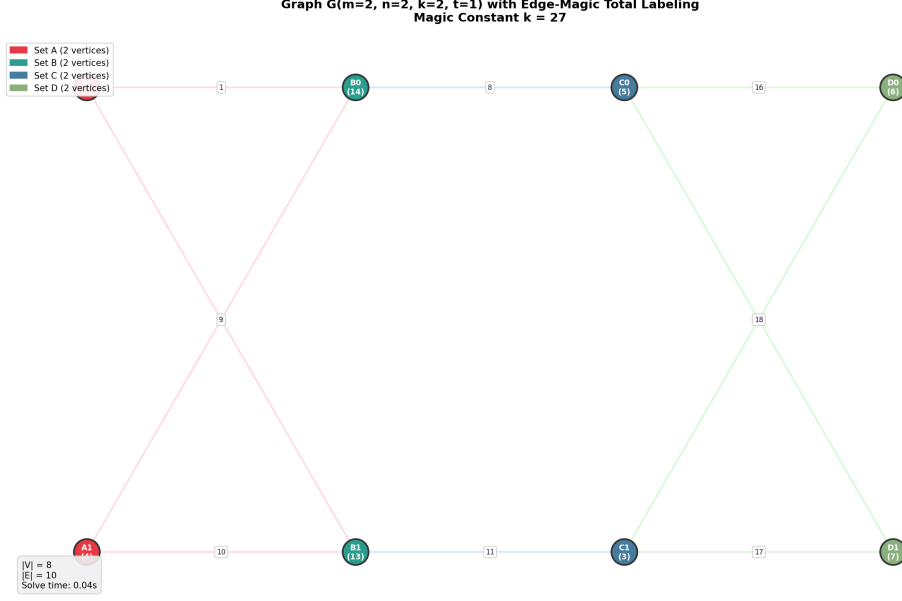


Figure 1: The graph $G(2, 2, 2, 1)$ with the four parts A, B, C, D (left to right), the two complete-bipartite layers $K_{2,2}$ ($A-B$ and $C-D$) and the 1-regular middle layer ($B-C$). Vertex and edge labels of a computed EMTL are shown; every edge sum equals the magic constant κ .

3.2 Circulant construction of the middle layer

The middle layer is built by the circulant rule

$$E_{BC} = \{ (B_i, C_{(i+j) \bmod n}) : i \in \{0, \dots, n-1\}, j \in \{0, \dots, t-1\} \}. \quad (3)$$

Proposition 1 (Regularity of the middle layer). *For $0 \leq t \leq n$, the bipartite graph $(B \cup C, E_{BC})$ of (3) is t -regular on both sides.*

Proof. By (3), each B_i is adjacent to exactly the t distinct vertices $C_{(i) \bmod n}, \dots, C_{(i+t-1) \bmod n}$, so B_i has degree t . Conversely, fix C_r . Then $(B_i, C_r) \in E_{BC}$ iff $r \equiv i + j \pmod{n}$ for some $j \in \{0, \dots, t-1\}$, i.e. $i \equiv r - j \pmod{n}$; these give the t distinct vertices $B_{(r) \bmod n}, \dots, B_{(r-t+1) \bmod n}$. Hence C_r also has degree t . $\square$

Proposition 2 (Degree sequence). *In $G(m, n, k, t)$ the degree is constant on each part:*

$$\deg(a) = n \ (a \in A), \quad \deg(b) = m + t \ (b \in B), \quad \deg(c) = k + t \ (c \in C), \quad \deg(d) = n \ (d \in D).$$

Proof. Each $a \in A$ is joined to all of B and nothing else, giving $\deg(a) = n$; symmetrically $\deg(d) = n$. Each $b \in B$ is joined to all of A (m edges) and, by Proposition 1, to t vertices of C , giving $\deg(b) = m + t$; symmetrically $\deg(c) = k + t$. $\square$

Figure 1 shows a small instance with the three layers and a found labeling.

4 Algebraic necessary conditions

This section records identities that every EMTL of G must satisfy. They are not needed to *solve* an instance, but they constrain the magic constant, supply the search bounds used by the solver, and act as independent post-solve checks.

Let f be an EMTL of $G = (V, E)$ with magic constant κ . Summing the magic equation of Definition 2 over all edges gives

$$\sum_{uv \in E} (f(u) + f(uv) + f(v)) = |E| \kappa. \quad (4)$$

Lemma 1 (Endpoint sum as a degree-weighted vertex sum). $\sum_{uv \in E} (f(u) + f(v)) = \sum_{v \in V} \deg(v) f(v)$.

Proof. Each label $f(v)$ contributes once to the sum for every edge incident with v , and there are exactly $\deg(v)$ such edges. $\square$

Proposition 3 (Degree-weighted identity). *Let $S_N := \sum_{i=1}^N i = \frac{N(N+1)}{2}$. Every EMTL of G satisfies*

$$|E| \kappa = S_N + \sum_{v \in V} (\deg(v) - 1) f(v). \quad (5)$$

Proof. Split (4) into edge- and vertex-contributions and apply Lemma 1:

$$|E| \kappa = \sum_{e \in E} f(e) + \sum_{v \in V} \deg(v) f(v).$$

Because f is a bijection onto $\{1, \dots, N\}$ we have $\sum_{e \in E} f(e) + \sum_{v \in V} f(v) = S_N$, hence $\sum_{e \in E} f(e) = S_N - \sum_{v \in V} f(v)$. Substituting yields (5). $\square$

Remark. For $G(m, n, k, t)$ the coefficient $\deg(v) - 1$ takes only the values $n - 1$ (on A and D), $m + t - 1$ (on B), and $k + t - 1$ (on C), by Proposition 2.

Corollary 1 (Divisibility / congruence condition). *With the notation of Proposition 3,*

$$\kappa = \frac{S_N + \sum_{v \in V} (\deg(v) - 1) f(v)}{|E|}, \quad \text{so} \quad S_N + \sum_{v \in V} (\deg(v) - 1) f(v) \equiv 0 \pmod{|E|}.$$

In particular, the numerator must be divisible by $|E|$. This is necessary but not sufficient for an EMTL to exist.

Proposition 4 (Universal bounds on the magic constant). *Every EMTL satisfies $6 \leq \kappa \leq 3N - 3$.*

Proof. Each edge sum is a sum of three *distinct* labels from $\{1, \dots, N\}$. The smallest such sum is $1 + 2 + 3 = 6$ and the largest is $N + (N - 1) + (N - 2) = 3N - 3$. $\square$

The implementation uses Proposition 4 as the domain of the magic-constant variable in the constraint model of Section 5.

5 Reduction and constraint model

5.1 The vertex-determination reduction

The magic equation is functional in the edge label: rearranging Definition 2 gives $f(uv) = \kappa - f(u) - f(v)$. This yields a clean reduction.

Proposition 5 (Vertex-determination). *Let $G = (V, E)$ and fix a constant κ and an injection $x: V \rightarrow \{1, \dots, N\}$. Define the induced edge values $y_{uv} := \kappa - x_u - x_v$ for $uv \in E$. Then (κ, x) extends to an EMTL of G if and only if*

- (a) $y_{uv} \in \{1, \dots, N\}$ for every $uv \in E$;
- (b) the values $\{y_{uv} : uv \in E\}$ are pairwise distinct; and
- (c) $\{y_{uv} : uv \in E\} \cap \{x_v : v \in V\} = \emptyset$.

When these hold, the EMTL is unique given (κ, x) , namely $f|_V = x$ and $f(uv) = y_{uv}$.

Proof. If f is an EMTL with $f|_V = x$ and magic constant κ , then the magic equation forces $f(uv) = \kappa - x_u - x_v = y_{uv}$, so the edge labels are determined; bijectivity of f is exactly the conjunction of (a)–(c) together with injectivity of x . Conversely, if (a)–(c) hold, then f defined by $f|_V = x$, $f(uv) = y_{uv}$ takes values in $\{1, \dots, N\}$, is injective on $V \cup E$ (vertices are distinct by hypothesis, edges by (b), and the two sets are disjoint by (c)), and therefore bijective since $|V \cup E| = N$. By construction it satisfies every magic equation. $\square$

Proposition 5 shows that EMTL existence is, in principle, a search over κ and the vertex labeling alone: the edge labels never need to be guessed. The solver below keeps explicit edge variables for clarity and for direct verification, but it exploits exactly this structure — fixing vertex labels forces edge labels, which then propagates strongly through the bijection constraint.

5.2 CP-SAT formulation

We model EMTL existence as a constraint-satisfaction problem (no objective). Let $N = |V| + |E|$.

Variables.

- a label variable $x_v \in \{1, \dots, N\}$ for each vertex $v \in V$;
- a label variable $x_e \in \{1, \dots, N\}$ for each edge $e \in E$;
- the magic-constant variable $\kappa \in \{6, \dots, 3N - 3\}$ (Proposition 4).

Constraints.

$$\text{AllDifferent}(\{x_v : v \in V\} \cup \{x_e : e \in E\}), \quad (\text{C1})$$

$$x_u + x_{uv} + x_v = \kappa \quad \text{for every } uv \in E. \quad (\text{C2})$$

Since the N label variables all share the domain $\{1, \dots, N\}$, constraint (C1) forces them to be a bijection onto $\{1, \dots, N\}$; (C2) is the magic property. The model therefore has $N + 1$ integer variables, one all-different constraint, and $|E|$ linear equations. By (1)–(2) the size grows quadratically in n because of the complete-bipartite layers.

5.3 Why CP-SAT

We solve the model with Google OR-Tools CP-SAT [8], a hybrid solver that combines constraint propagation over integer domains with SAT-style conflict-driven clause learning. Two features of the EMTL model make it a good fit. First, by Proposition 5 each magic equation (C2) is functional, so as soon as two of the three labels on an edge are fixed the third is forced; propagation thus turns partial vertex assignments into forced edge labels. Second, the global all-different constraint (C1) couples every label, so a single forced value can prune many domains, and learned conflict clauses prevent the search from rediscovering the same infeasible patterns — of which there are many, since the family contains many structurally identical sub-configurations. When the search exhausts the tree without a solution, the solver returns a genuine proof of infeasibility rather than a mere timeout.

6 Reference implementation

The solver is implemented in Python (`emtl_solver.py`) on top of `networkx` and OR-Tools, with a `matplotlib` visualizer and a Streamlit web interface. Table 1 maps the mathematical objects of this paper to the code. The end-to-end entry point `solve_emtl(m, n, k, t, timeout)` validates the parameters, constructs $G(m, n, k, t)$, builds and solves the model of Section 5, and independently verifies any returned labeling against Definition 2.

Mathematical object	Code
Parameters (m, n, k, t) , counts $ V , E , N$	<code>GraphParameters</code>
Graph $G(m, n, k, t)$ and circulant layer (3)	<code>GraphConstructor.construct</code>
Variables, (C1)–(C2), solve	<code>EMTLSolver.solve</code>
Bijection + magic check (Def. 2)	<code>EMTLSolver.verify_labeling</code>
Drawing of a labeled instance	<code>EMTLVisualizer.visualize</code>
Full pipeline	<code>solve_emtl</code>

Table 1: Correspondence between the formal model and the reference implementation.

m	n	k	t	$ V $	$ E $	N	Outcome	Time (s)
1	1	1	0	4	2	6	infeasible (no EMTL)	0.03
1	2	1	1	6	6	12	$\kappa = 17$	0.02
2	2	2	0	8	8	16	$\kappa = 26$	0.03
2	2	2	1	8	10	18	$\kappa = 27$	0.03
2	3	2	2	10	18	28	$\kappa = 33$	0.09
3	3	3	3	12	27	39	$\kappa = 50$	0.34

Table 2: Representative EMTL instances on $G(m, n, k, t)$ with the magic constant found by CP-SAT (or a proof that none exists). Every labeling reported here was independently re-verified against Definition 2.

7 Experiments

Setup. All runs use OR-Tools CP-SAT with the bounds of Proposition 4, a per-instance time limit, and the solver’s default portfolio of search workers. Reported times are wall-clock seconds on a commodity laptop and are indicative rather than benchmark-grade; the qualitative point is that small and moderate instances of $G(m, n, k, t)$ are decided quickly and exactly. For every instance reported as FOUND, the returned labeling was re-checked against Definition 2 by an independent verifier.

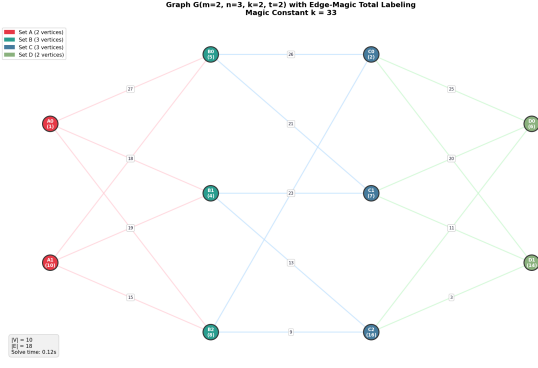
Results. Table 2 summarizes representative runs; every instance was decided in under half a second, and Figure 2 shows two of the computed labelings.

The smallest instance, $G(1, 1, 1, 0)$, is *infeasible*, and this is predicted by the theory of Section 4: all four vertices have degree 1, so every coefficient $\deg(v) - 1$ in (5) vanishes and the identity reduces to $|E|\kappa = S_N = 21$ with $|E| = 2$. Since 21 is not divisible by 2, Corollary 1 already rules out any EMTL; the solver’s infeasibility certificate matches this prediction.

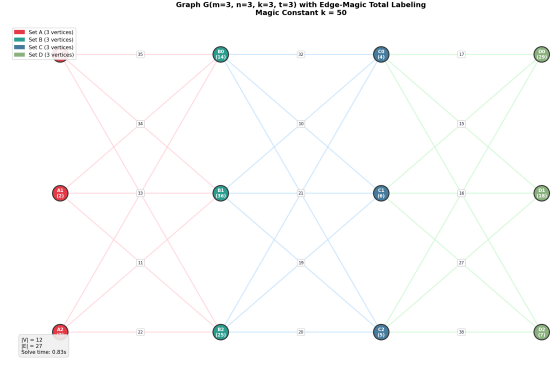
A worked example. For $(m, n, k, t) = (2, 2, 2, 1)$ we have $|V| = 8$, $|E| = 10$, and $N = 18$. The solver finds an EMTL with magic constant $\kappa = 27$; one valid labeling is shown in Figure 1, and `verify_labeling` confirms that all ten edge sums equal 27 and that the eighteen labels form a bijection onto $\{1, \dots, 18\}$.

8 Conclusion

We studied edge-magic total labelings on a structured four-partite family $G(m, n, k, t)$. We derived a degree-weighted identity, a divisibility condition, and bounds on the magic constant, and proved a vertex-determination reduction that pins down the labeling once the vertex labels and κ are fixed. Casting existence as a constraint-satisfaction problem and solving it with CP-SAT gives an exact method that returns a verified labeling, an infeasibility proof, or a timeout. Natural next steps are to characterize the parameter regions in which $G(m, n, k, t)$



(a) $G(2, 3, 2, 2)$



(b) $G(3, 3, 3, 3)$

Figure 2: Computed edge-magic total labelings for two larger members of the family. Parts are colored A, B, C, D from left to right; each edge label equals the magic constant minus its two endpoint labels.

is edge-magic, to add symmetry-breaking constraints exploiting the circulant structure of the middle layer, and to push the solver to larger instances.

Availability. The source code, test suite, visualizer, and this article are available in the project repository.

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